

Perturb2Target

From a genome-scale T-cell screen to directional,
genetics-anchored, druggable targets

**Screens tell you which genes matter.
We tell you which way to push them — and whether you can drug them.**

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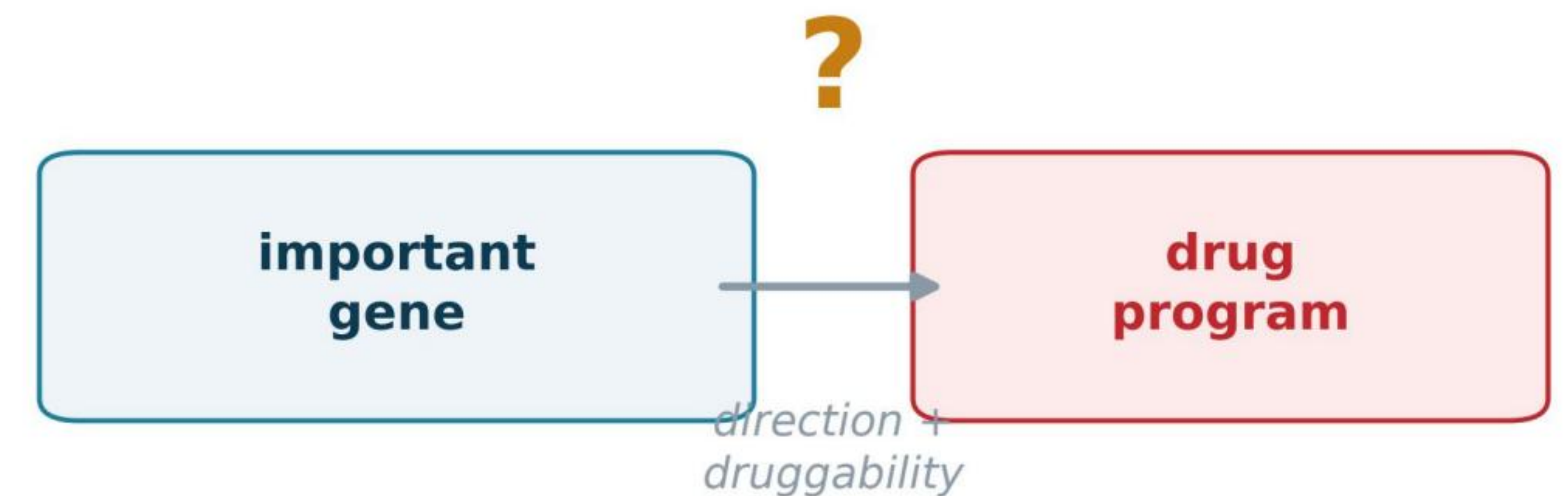
CD4⁺ T-cell CRISPRi Perturb-seq · 1,923 directional nominations · 6 novel leads

3-minute pitch

THE PROBLEM

The gap: screens rank genes, not what to do about them

- A genome-scale Perturb-seq screen tells you a gene is important.
- It does NOT tell you the therapeutic action:
 - **agonize a brake?** or **antagonize a driver?**
- Genetics (GWAS) anchors which genes, but is direction-agnostic.
- Most methods rank by network centrality — which rewards how well-studied a gene is, not what a drug program needs.



WHAT WE BUILT

The method: a directional score with an empirical null

1. Directional program score

signed sum over a 48-gene inflammatory signature —
positive = pushes program up, negative = down

2. Empirical null

500 random signed gene-sets per condition → z-score + FDR,
so the direction call is calibrated, not asserted

3. Four-layer integration

causal (0.34) · genetics (0.30) · druggability (0.22) · novelty (0.14)
— direction kept as annotation, never folded into the score

Output

1,923

directional nominations

1,290 antagonize drivers

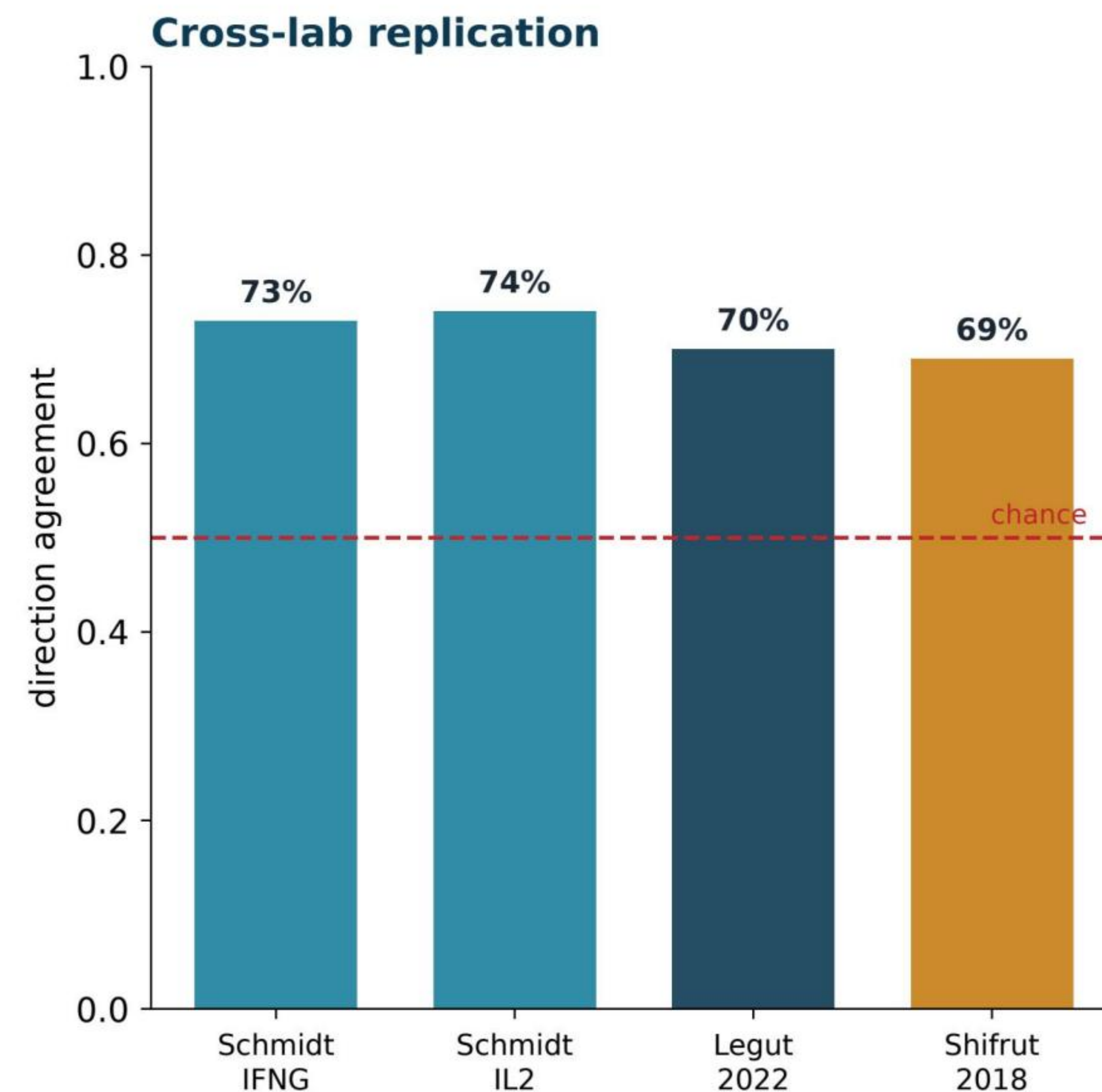
633 agonize brakes

*each with genetics +
druggability + novelty*

DOES IT HOLD UP?

The direction call is real — and it replicates across labs

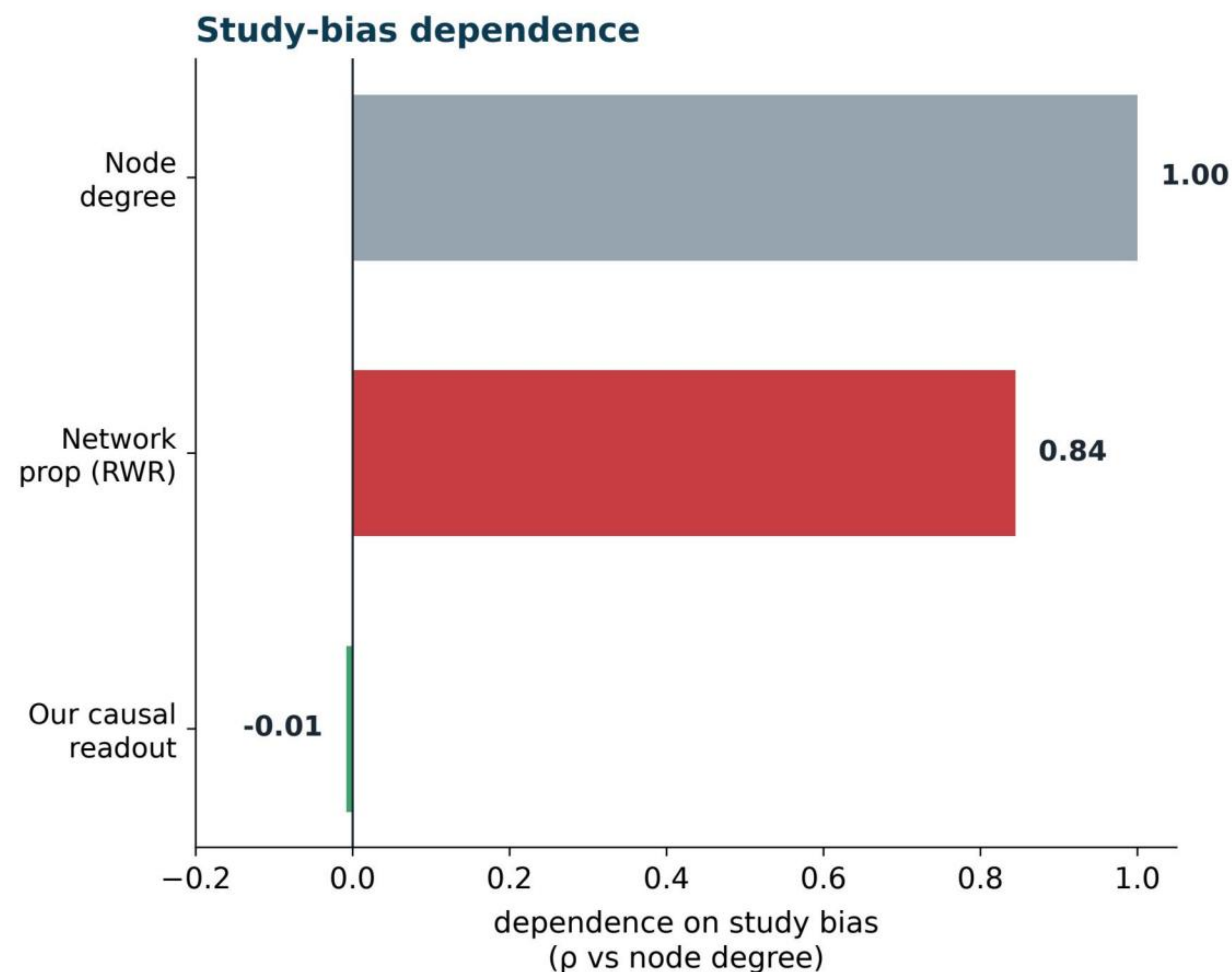
- Matches approved-drug mechanism in 73% of cases (ChEMBL gold standard; $p = 2 \times 10^{-5}$).
- Recovers known targets with the RIGHT direction: IL4R, IL2RA, STAT3, IL6R at the top.
- Replicates in 3 independent labs (Schmidt, Legut, Shifrut): $\rho = 0.49$ – 0.52 , direction agreement 69–74%.
- Honest about limits: on the underpowered agonize arm (6 targets) we report the negative result, not a spun win.



THE STANDOUT CLAIM

Why we're different: everyone else ranks by how well-studied a gene is

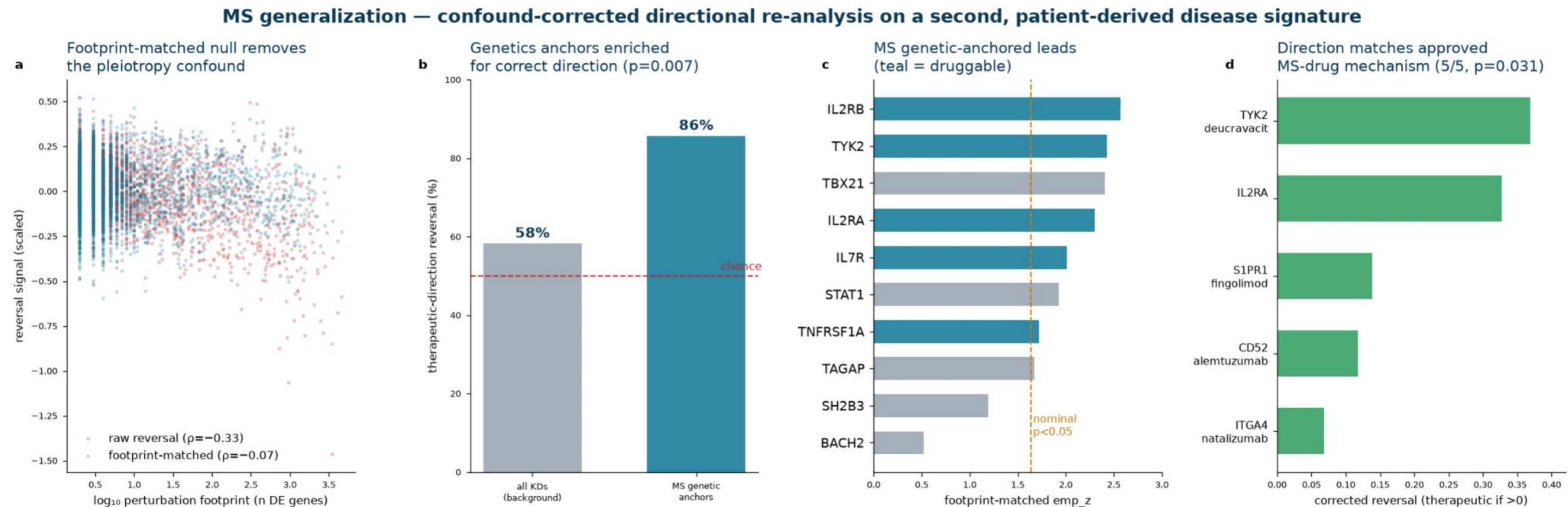
- Network-propagation methods track node degree almost perfectly ($\rho = 0.84$ with study bias) — they recover hubs.
- Our causal, directional readout is study-bias-independent ($\rho = -0.01$): it measures what a drug program needs.
- We measure direction — the property GWAS and centrality cannot supply — and show it survives the confound.



AND IT GENERALIZES

The whole thesis, reproduced on a second disease (multiple sclerosis)

- We re-ran the entire pipeline — unchanged — on a patient-derived MS signature (1.9M CD4 cells, two cohorts).
- **Naive reversal fails exactly as we predict: uncorrected, its top hits are pleiotropic housekeeping hubs.**
- **The signal lives at genetics × direction: MS genetic anchors are 86% correctly-directed vs 58% ($p = 0.007$)...**
- **...and the direction matches approved-MS-drug mechanism in 5 / 5 measurable targets ($p = 0.031$).**



Same directional score · empirical null · genetics layer · druggability filter — MS is one worked example of any CD4-T-cell disease.

THE PAYOFF

Six novel, druggable, genetics-anchored leads — with one falsifiable bet

Gene	Direction	Genetics	Class
SIK2	agonize	asthma	kinase
PTPN2	agonize	RA	phosphatase
STAT6	agonize	hyper-IgE	SM
CTSH	antagonize	T1D	cathepsin
STAT5B	antagonize	immune-dysreg	SM
NEK6	antagonize	asthma	kinase

Flagship falsifiable bet

SIK2 (brake, asthma):

CRISPRi raises the program ($z = +4.7$). So CRISPRa should LOWER it in resting CD4⁺ T cells.

Falsified if $\Delta \geq 0$ at $n=8/\text{arm}$.

Panel of 6 · ~96 conditions ·
~3-4 mo · ~\$60-90k · pre-registered

A reusable method, not just a hit-list

- The method is the product: directional_screen + Path2Drug are dataset-agnostic — any perturbation × readout screen, any cell type.
- Path2Drug reaches undruggable TFs by routing to a druggable node on the pathway — grounded in the atlas, not a curated graph.
- Everything open + reproducible: pipeline, benchmark, interactive explorer, and a pre-registered validation protocol.